PROJECT DESIGN PHASE-II

Solution Requirements (Functional & Non-functional)

Date	28 October 2026	**Maximum Marks**	4 Marks
Team ID	EN25CS3011166	**Project Name**	Health360: Cardiovascular Risk Analytics

Functional Requirements:

Following are the functional requirements of the proposed business intelligence and clinical analytics solution.

FR No.	Functional Requirement (Epic)	Sub Requirement (Story / Sub-Task)	Priority
1	**Data Access & Integration**	Integrate medical records from `smart_healthcare_dataset` containing metrics like Age (19-79), BMI, Cholesterol, BP, Glucose, Exercise Level, Fatigue, and Chest Pain.	**High**
2	**Data Processing**	Prepare clinical data, categorize Blood Pressure levels (113.40-119.07+), calculate AVG(Heart Disease), and generate Patient Risk Heatmaps by BMI and BP category.	**High**
3	**Business Analysis**	Compute core analytical measures: Age-Based Cardiac Risk Trends, Fatigue v/s Stroke Risk, Exercise Level vs Risk, Chest Pain v/s Dizziness, and Alcohol/Smoking Risk score.	**High**
4	**Business Insights & Analytical Outputs**	Deliver unified interactive dashboard view displaying 8 diagnostic charts: Age trends, Risk Heatmap, Alcohol & Smoking, Cholesterol level, BP risk, BMI risk, Fatigue, and Glucose risk.	**High**
5	**User Interaction & Analysis**	Provide interactive filtering across parameters (BMI category, Exercise level 0/1/2, Age group) and detailed visual tooltips with average percentage breakdowns.	**Medium**
6	**Data Update & Reporting**	Support dynamic updates from source tables (48 marks, record count tracking) and export summarized clinical cohort risk reports for medical care teams.	**Medium**

Non-functional Requirements:

Following are the non-functional requirements of the proposed analytical solution.

NFR No.	Non-Functional Requirement	Description
NFR-1	**Usability**	Dashboard must maintain clean tabular layout, clear axis labeling (0.00% to 50.00%), and intuitive chart headers across desktop and tablet views.
NFR-2	**Security**	Implement patient data anonymization (HIPAA compliant) and role-based access for clinical analysts viewing heart disease and stroke statistics.
NFR-3	**Reliability**	Ensure accurate calculation of aggregate measures (AVG Heart Disease, Health Risk Score) across varying sample sizes without drift.
NFR-4	**Performance**	Dashboard elements must render dynamically in under 2 seconds during filter applications in Tableau Web / Streamlit.
NFR-5	**Availability**	Provide 99.5% uptime for web dashboard hosting on Tableau Public / enterprise BI server for uninterrupted clinical review.
NFR-6	**Scalability**	Architecture must support scaling from initial dataset batches (48 marks) up to millions of real-time EHR patient records seamlessly.